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# Designing Trust in Autonomous Aerial Mobility

From technical safety to perceived safety: the role of transparency and UX in trust calibration through the Safety-Transparency Trust Framework

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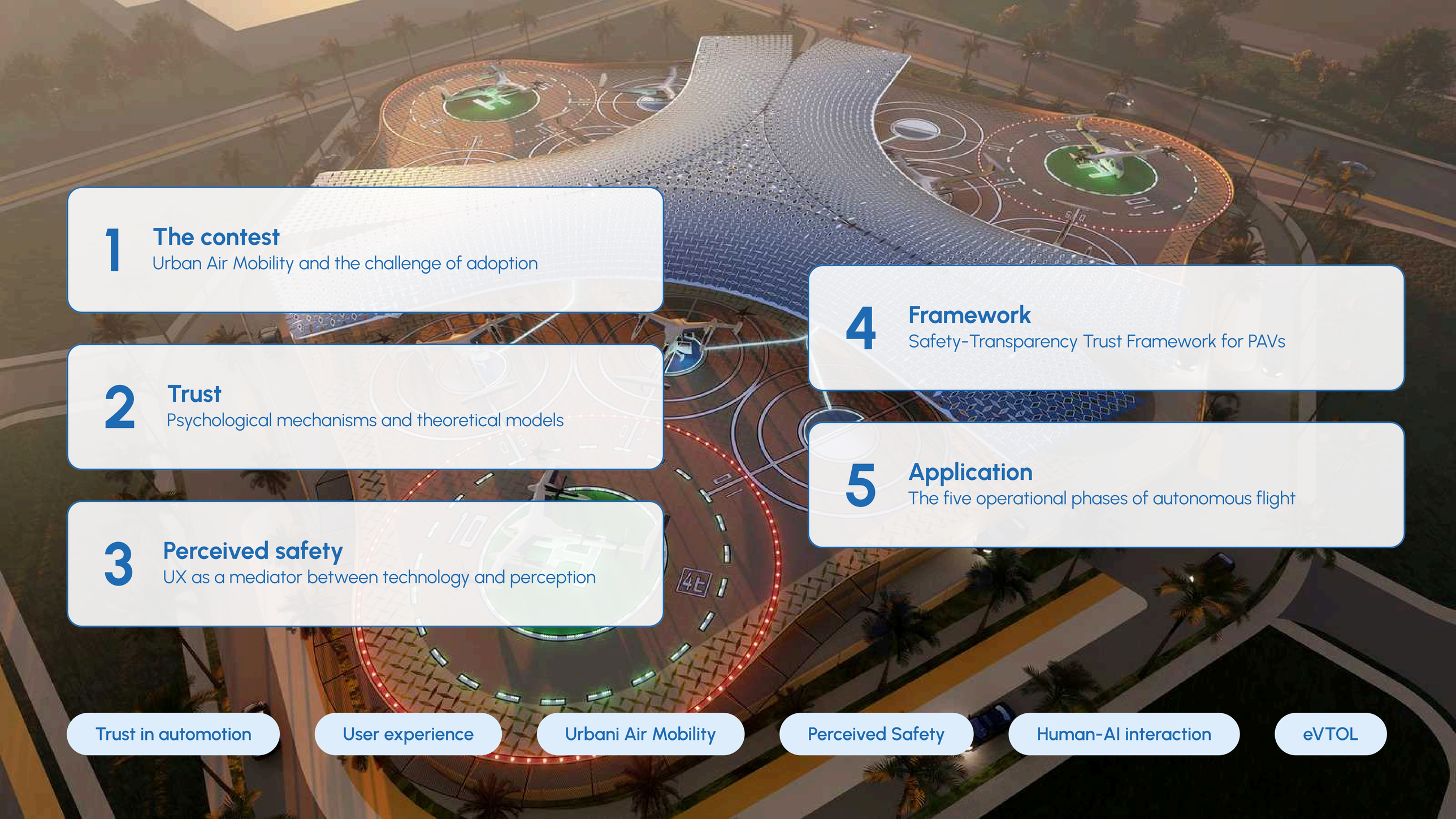
# Design trust in autonomous aerial mobility

Dalla sicurezza tecnica alla sicurezza percepita, il ruolo della trasparenza e della UX nella calibrazione della fiducia attraverso il Safety-Transparency Trust Framework.

Politecnico di Milano, School of Design
MSc in Digital&Interaction Design

Master's Degree Thesis
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Thesis supervisor
Professor Maximiliano Romero



**1 The contest**
Urban Air Mobility and the challenge of adoption

**2 Trust**
Psychological mechanisms and theoretical models

**3 Perceived safety**
UX as a mediator between technology and perception

**4 Framework**
Safety-Transparency Trust Framework for PAVs

**5 Application**
The five operational phases of autonomous flight

Trust in automotion | User experience | Urbani Air Mobility | Perceived Safety | Human-AI interaction | eVTOL

Scenario

# The Evolution of Personal Flight



The concept of flying cars has long fueled the collective imagination, embodying aspirations for technological progress. Reinforced by the entertainment industry, which has depicted personal flight in numerous fictional works like Back to the Future, helping to transform a visionary idea into a shared technological goal.

Scenario

# A paradigm shift in Urban Air Mobility




Urban Air Mobility (UAM) is an urban air transport system based on electric Vertical Take-Off and Landing (eVTOL) aircraft. It is designed for the transport of passengers and cargo, including autonomous modes, across direct and intra-city routes within metropolitan areas.

→

## eVTOL

Passenger &
Cargo Transport

## Dynamic Network

Hub and Spoke,
Point to Points

Scenario

# eVTOL Configurations



## Vectored Thrust

Tilting motors for vertical and horizontal thrust: maximum efficiency over long distances, but high technical complexity.

Passenger transport: point-to-point connections within metropolitan areas, urban centers, and airports.

Range: 200+ km

Complexity: High



## Lift + Cruise

Dedicated rotors for vertical flight and independent motors for forward thrust. An excellent compromise between construction and certification.

Cargo transport: between ware-houses and urban distribution hubs.

Certification: simplified

Complexity: Moderate



## Wingless

No fixed wings, continuous rotor thrust. Ideal for short urban distances, ensuring high redundancy and simplicity

Transport of passengers, cargo, medical personnel, equipment, organs, or patients.

Range: Urban short

Complexity: Low

Scenario

# Actors









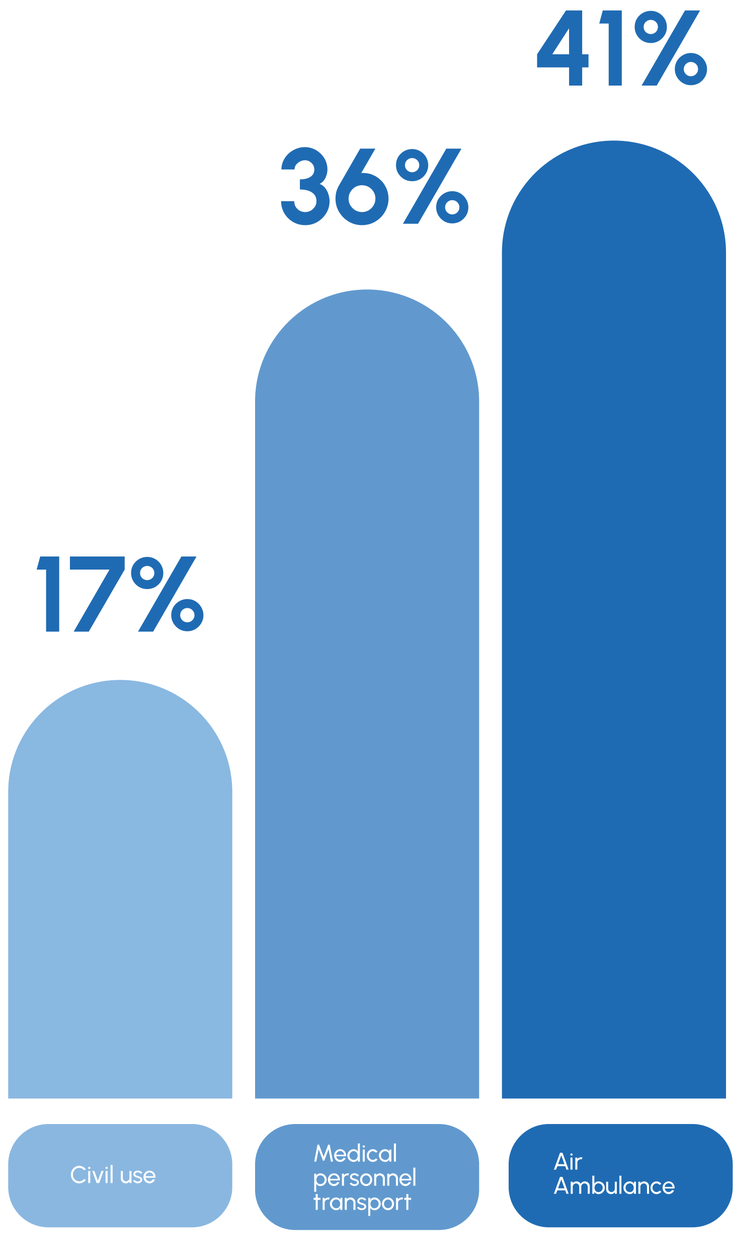



| Joby Aviation | Archer Aviation | Volo copter | Wisk Aero | E-Hang |
|---|---|---|---|---|
| Joby S4 | Midnight | VoloCity | Gen 6 | EH216-S |
| USA | USA | Germany | USA | Cina |
| 4 pax | 4 pax | 1 pax | 4 pax | 2 pax |
| Automation assistent (L3-L4) | Automation assistent (L3-L4) | Automation assistent (L3) | Full Automation (L4 - L5) | Full Automation (L4 - L5) |

# The technology is ready for take-off; are the people?

Scenario

# Statistic




Qualitative interviews suggest that this discrepancy stems from the perception of civil air taxis as a potentially elitary and expensive service, which is seen as less socially justifiable than a life-saving mission.

EASA & Airbus, survey of 4,000 European citizens, 2021

Scenario

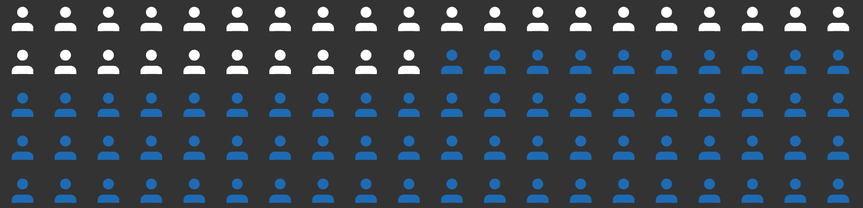
# Statistic

The issue isn't technical, it's trust

70% accept a piloted air taxi

**70%**

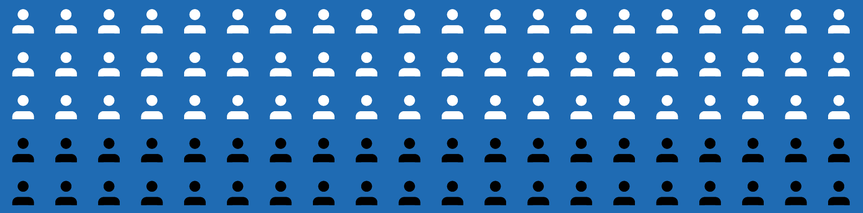
Onboard pilot



40% acceptance rate for autonomous flight

**44%**

Autonomous Pilot



Public acceptance of UAM is not determined solely by engineering parameters; rather, it emerges from a complex ecosystem of social factors, encompassing psychological, environmental, and cultural dimensions.

EASA & Airbus, survey of 4,000 European citizens, 2021

Scenario

# The invisible barrier

**Technical Safety**

Objective and actual system reliability

- GNSS/IMU Navigation
- LiDAR and Visioan-Based Sensors
- System Redundancy
- EASA/FAA Certifications

Design gap

The user doesn't see the math

**Perspective Safety**

User's subjective and psychological evaluation

- No Pilot - Loss of Agency
- Opaque AI - Black Box Effect
- Cognitive Biases - Risk Distortion
- Close Environment - No Way Out

Scenario

# Project goals

## 01.

Understand the **psychological mechanisms and biases** that govern trust formation in autonomous flight systems.

## 02.

Identify **design guidelines** and **tools** that help bridge the gap between technical safety and perceived safety.

## 03.

Promote **Appropriate Reliance**, enabling the user to calibrate their intervention based on the system's actual capabilities and operational limits.

# What psychological mechanisms regulate trust in autonomous systems?

Analysis

# Trust in interpersonal relation

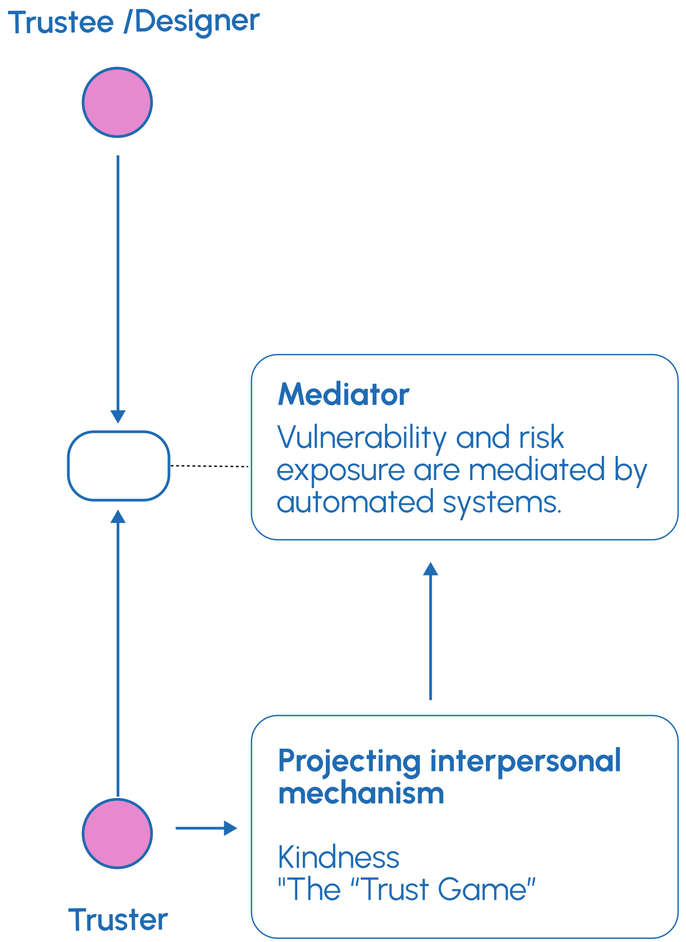



"The willingness of a party to be vulnerable to the actions of another party based on the expectation that the truster, irrespective of the ability to monitor or controll that other party"

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. Academy of Management Review, 20(3)

Analysis

# Trust in automation




"The attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability"

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors thatinfluence trust. Human Factors,

Analysis

# The Mechanisms of Trust

**Interpersonal trust**

Ability

Integrity

Benevolence

**I trust your intentions, even if I don't control you.**

*"I trust you because you are capable, honest, and have my best interests at heart"*

Moral Connection

**Trust in automation**

Performance

Process

Purpose

**I trust your capabilities to achieve my goal.**

*"I trust the system because it works, I understand how it operates, and I know what it was built for."*

Logical Connection

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. Academy of Management Review, 20(3)
Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. Human Factors, 46(1),

While trust in interpersonal relationships tends to grow slowly, machines often elicit a high level of initial perceived trust. However, this can collapse drastically at the first error, making the calibration of trust extremely fragile.

Analysis

# Model in Trust automation

| Autor | Theory | Direction |
| --- | --- | --- |
| **Lee & See 2004** | They developed a dynamic model to explain why users may over-rely on automation (misuse) or unjustifiably reject it (disuse). This framework identifies three fundamental pillars and three calibration parameters, enabling the user to build trust and achieve 'Appropriate Reliance' on the system. | **The design must transparently communicate the Operational Design Domain (ODD) limits, allowing the user to transition from "blind faith" to a trust rooted in the knowledge of the vehicle's actual processes and performance.** |
| **Parasuraman et al. 2000** | The model distinguishes four cognitive functions that can be automated. For each, there is a scale ranging from 1 (manual) to 10 (fully autonomous). The key point is that the level of automation (LoA) should not be fixed, but rather adapt dynamically to situational demands | **These systems change the level of automation based on how tired or busy the pilot is. This helps them stay focused without having too much work.** |
| **Hoff & Bashir 2015** | The model distinguishes between three overlapping levels of trust: Dispositional Trust, Situational Trust, and Learned Trust. Transparency is the enabling factor for the transition from situational trust to learned trust: when the system's logic is understandable, calibration becomes effective. | **Trust grows step by step over time. It is not enough if the system works well; the user must also understand how it work** |

Analysis

# Why the trust calibration is important?



## Over trust

The user perceives the machine as more reliable or intelligent than it truly is.

The Airbus A320 Crash (1988)



## Under trust

The user's trust is lower than the system's actual capabilities. Even if the aircraft is technically ultra-safe, the user does not trust it.

Costa concordia, 2012

Good calibration is dinamic process refers to the correspondence between a person's trust in automation and the automation's actual capability.

Analysis

200+ Bias Tipologies

# Cognitive Bias

Cognitive bias are involontary “filter”, mental shortcuts that simplify complex decisions, producing systematic distortions perceived risk, especially under conditions of uncertainty and when the technology is unfamiliar or highly innovative.

**Automation bias**

Excessive delegation to the system, even when it fails.

**Availability heuristic**

Events are evaluated based on relevant examples that are easily recalled from memory.

**Status quo bias**

Cognitive Bias, Drives the user to maintain familiar autonomous modes

**Confirmation bias**

Tendency to select and interpret information in a way that confirms one's pre-existing beliefs.

Analysis

# Cognitive Bias

200+ Bias Tipologies

### Action bias

Preference for action over inaction, especially in situations of stress or uncertainty.

### Framing effect

Decisions are influenced by the way information is presented.

### Illusory control

A universal tendency to overestimate one's own ability to influence events that are determined by chance.

### Salience bias

The tendency to give more weight to information that is more prominent, unusual, or visually striking.

# How can we help the system and the passenger work together?

Analysis

# The user as the place where trust manifests

Trust is not a property of the system. it is a dynamic construct that takes shape through the interaction between the user, technology, and the operational context.

**Absence of a human pilot**

The symbolic security reference is missing. The guarantor has disappeared.

**High perceived flight vulnerability**

Closed, three-dimensional, uninterruptible environment, no escape route

**Decision-making opacity of artificial intelligence**

AI makes decisions without explaining them, an endemic black box effect.

Analysis

# Key research insights

20 pappers

Absence of a human pilot and loss of user agency

| Findings | Insights | Implication |
| --- | --- | --- |
| Participants expressed concern about sexual assault and passenger aggression in an enclosed automated space without crew. Many refused to consider any form of shared autonomous mobility for this reason. | The discomfort does not stem from technical failure, but from the absence of a human authority figure capable of managing the unexpected | **The pilot is also a social protector, not just a technical one** |
| | | Booz Allen Hamilton (2018) |
| When the vehicle had a name (Iris), a female voice, and a human identity, users showed a lower heart rate and a reduced startle response during critical events, compared to a silent and identity-less vehicle | A humanized interface partially replaces the psychological function of the pilot | **Anthropomorphism is a legitimate and measurable design tool, not merely an aesthetic choice** |
| | | Waytz et al. (2014) |
| Under real risk (at an altitude of 70m), users increased their monitoring even when they reported trusting the system. The enclosed flight environment amplifies the sense of helplessness. | Stated trust and behavioral trust do not align under stress. | **We only let the machine decide when the risks are very low. If the consequences are big, we want to stay in control** |
| | | Hoesterey&Onnasch (2022) |

| | Findings | Insights | Implication |
|---|---|---|---|
| Transparency and Explainable AI | Even wrong explanations increase trust. A system that explains its behavior, even incorrectly, is perceived as more credible than a silent one. | Credibility depends not on accuracy, but on the presence of an explanation | **Poorly designed transparency can reinforce misinformation and overtrust** |
| | | | Manger et al. (2025) |
| | When pilots could see and negotiate the system's decision weights (sliders), trust improved and cognitive load decreased. Without explanations, distrust and uncertainty increased. | Legible autonomy is collaborative autonomy | **Providing the user with visibility into the system's decision-making parameters, even partially, produces more calibrated reliance** |
| | | | Battiste et al. (2018) |
| Perceived control and calibration | Contextual explanations (why the car is braking, why it is swerving) reduce anxiety and increase automation acceptance, even when users had no possibility of intervening | Understanding replaces the need for physical control | **Communicating the "why" behind every maneuver is more effective than offering an override button** |
| | | | Koo et al. (2015) |
| | Perceived control increases significantly only when the system makes its own limits explicit. Level 7 automation (acting autonomously but constantly informing the user) produces better-calibrated trust than Level 10 (acting in silence). | It is not about returning control, it is about making the limits visible | **To feel in control, users don't need to drive. They just need clear information about what the system can and cannot do** |
| | | | Walker et al.(2023) |

# How can UX transform technical security into perceived security?

Framework

# UX as a Facilitator

## UX as a Narrator

Builds the social legitimacy of the UAM system. It utilizes anthropomorphism to create an emotional bond that increases error tolerance. It communicates public benefits to broaden societal acceptance

Adopting Anthropomorphism

**Technical Safety** ⟷ **Perceived Safety**

## UX as a Translator

Converts opaque AI outputs into readable "signals". It translates technical variables into meaningful insights for the non-expert user. The interface is a medium through which trust evolves.

Interface as a conversational partner

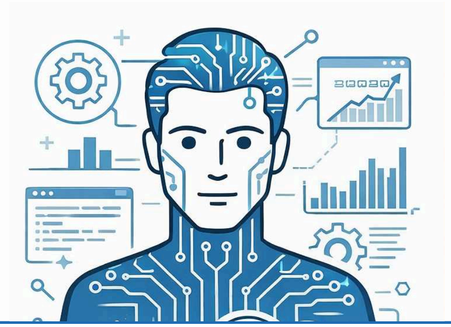
## UX as a Regulator

Manages the dynamic equilibrium between overtrust and undertrust. It maintains the user in the "cognitive loop". It provides perceived behavioral control, the expectation of being able to regain control whenever necessary.

Calibration throught information

Framework

# Calibration Archetypes



## The Analytical verifier

*Show me the data.*

Cognitive Trust

Need for procedural logic and consistency

Risk of distrust if data is perceived as insufficient



## The Vulnerable passenger

*Am I safe here?*

Emotional Trust

*Risk sensitivity and emotional vulnerability*

*Unmotivated takeover, attempted intervention for situational anxiety*



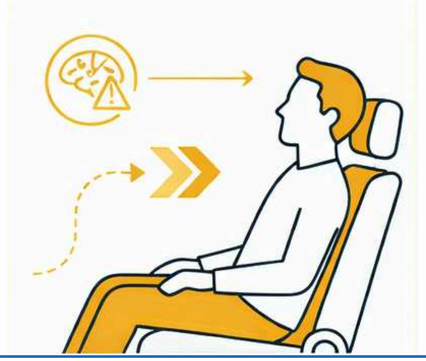
## The Control seeking

*I want to know what's happening.*

*Operational Trust*

*Need for active supervision and control*

*Human-machine conflict, Illusory control, and trust oscillation*



## The Delegating Optimist

*I trust technology.*

Analogical Trust

Tendency to delegate based on past experience

Overreliance and loss of situational awareness

Framework

# Calibration Archetypes

### Takeover feeling

Is a common thread among the rational, the anxious, and the supervisor (though for different reasons). The trusting user, however, surrenders their agency, risking a cognitive loop.

### Predictability mental model

Each profile brings a pre-existing mental model (airplane, elevator, car). The discrepancy between expectations and the system's behavior is the primary trigger for distrust

### Necessary transparency

Everyone reacts to transparency, but with opposing needs: too much overloads the rational user and frightens the anxious one; too little loses the supervisor and induces a false sense of calm in the trusting one.

### Not Calibration trust

No profile exhibits proportionate trust: the rational and the anxious: undertrust; the trusting: overtrust; the supervisor: continuous oscillation.

How do you design for trust when the passenger cannot see, touch, or question the system that is carrying them?

Framework

# Paradigm shift

## Maximization of trust

- Inducing the user to trust the system regardless of the circumstances" o "Coaxing the user into unconditional trust
- Overtrust: blind reliance and automation bias
- Reduction of human vigilance" o "Decreased human oversigh
- Risk of underestimating technical limitations

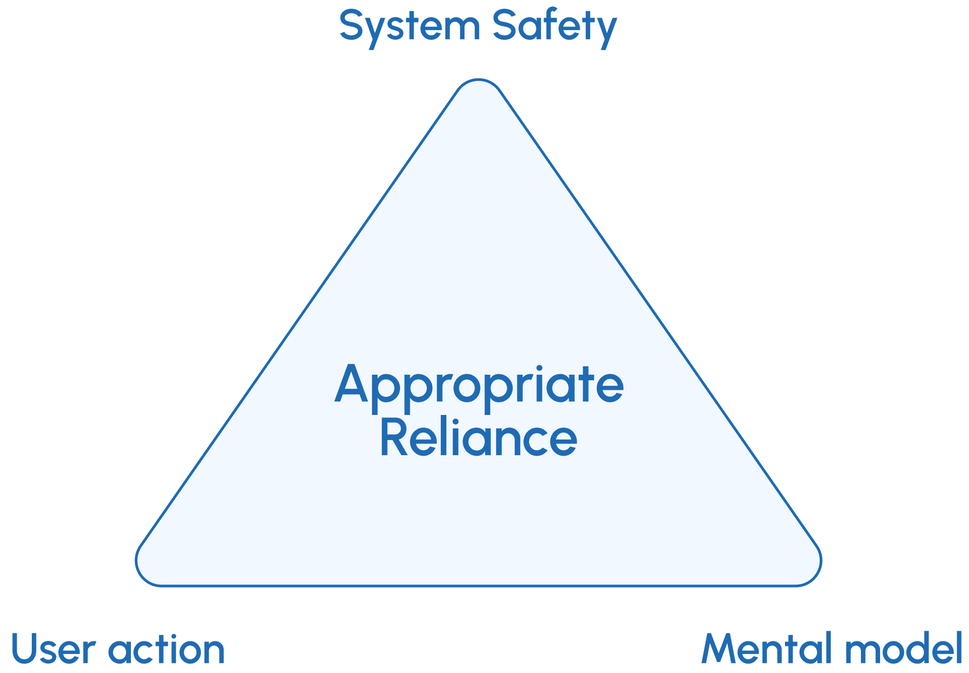
## Trust calibrated

- Aligning trust with the system's actual capabilities
- Aims for the optimal balance
- Transparent and informative
- Appropriate reliance and active monitoring

Framework

# Safety-Trasparency Trust Framework

A conceptual design model that dynamically calibrates user trust in autonomous flight systems, aligning the aircraft's actual operational capabilities with the passenger's mental model through the mediation of UX and AI transparency.



Framework

# Strategic Axes

It is organized into five macro-directive design axes, keeping the user as a vigilant partner of the system, transforming autonomy into a form of structural cooperation between man and machine.

Axe 1

### Emotional Risk Meditation & Explanability

Transform "fear of the unknow" into "understanding the maneuver".

Axe 2

### Adaptive Trust Calibration

Create an "elastic" interface that adapts based on who is on board.

Axe 3

### Active supervision & Cognitve Visibility

Eliminate the "Black Box" effect. AI must be a readable partner.

Axe 4

### Active Supervision & Anti-automation bias

Prevent the user from falling asleep or becoming overly trusting ( complacency).

Axe 5

### Agency Reconstruction & Longitudinal Trust

Make the user feel in control and build trust over time.

Framework

# Logical Architecture

## Cognitive dimension

How the user interprets risk, processes information and builds their mental model.

- Axe 1
- Axe 2

## Regulatory dimension

Trust as a variable to be calibrated according to actual capabilities and operating conditions.

- Axe 3
- Axe 4

## Evolutionary dimension

Imanaging trust over the long term, preventing the erosion of skills, rebuilding agency over time.

- Axe 5

# How does the system interaction change during the different phases of flight?

Framework

# Elastic interface

A single interface that expands based on control needs to avoid cognite vulnerability

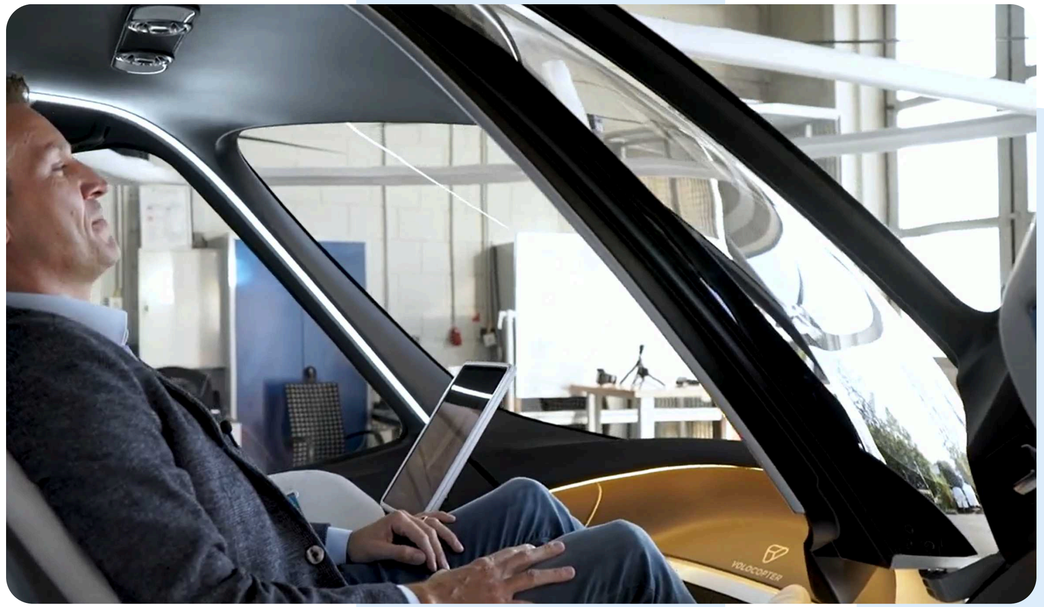
Surface Layer (Common elements)

*Anxios, Trusting*

Deep Layer (specific elements)

*Rational, Supervisor*

VoloCity eVTOL






E-hang 184 eVTOL

Framework

# Preflight&Onboarding

The goal of this phase is to align the user's expectations with the technological reality of the aircraft, promoting the construction of a coherent mental model.

**Analytical Verifier / Control Seeking**
*Rational/Supervisor*
Loss of Awareness, Operational limits, When to intervene

Out-of-the-loop

**Vulnerable Passenger**
*Anxious*
Emotional Stability, New Perceived Reference, Negative Imagery

Availability Heuristic

**Delegating Optimist**
*Trusting*
Familiarity, Cognitively Active, Overtrust Risks

Complacency

## Interface Convergence

COMMON ELEMENT, ALL PROFILES
**Trajectory Preview Place**
Place, map, route, altitude, etc.

COMMON ELEMENT, ALL PROFILES
**Situational Micro-narratives**
Language as a design decision

**Profile System state + Explicit ODD limits**
Battery, status, weather, when the system can fail

**Introductory video, spatial audio**
Replaces vivid negative flight imagery with positive sensory experiences

**CTA + Vigilance micro-signals**
Familiarity, subtle anti-complacency signal

**UX Solution**
*Introducing an "onboarding process" that is not merely procedural, but informative, educational, and keeps the user cognitively engaged*

**Active Axes**

**_Emotional Risk Mediation**
*Primary, video micro-narratives*

**_Adaptive Trust Calibration**
*Secondary, elastic interface*

Framework

# Take-off & Climb

The take-off and climb phase represents the most emotionally and cognitively charged moment, as it consides with liftoff.

**Analytical Verifier / Control Seeking**
*Rational/Supervisor*
Telematic Interface: Why This Choice, Prevent Action Bias

Action bias

**Vulnerable Passenger**
*Anxious*
Redundancy Confirmation, Active Monitoring, Attitude Vs. Action.

Confirmation bias

**Delegating Optimist**
*Trusting*
Familiarity, False Calm Risk

Complacency

## Interface Convergence

COMMON ELEMENT, ALL PROFILES
**Sensory synchrony**
Physical-Digital Coherence

COMMON ELEMENT, ALL PROFILES
**Informational countdown**
Feedforward: Next-phase anticipation

**Simplified Telemetry + Motivation**
The "Why" behind that decision

**Confirmation messages**
Resolving the attitude-behavior discrepancy under stress.

**Micro-signals of vigilance,**
Comfort and continuity, and subtle "active system" cues to prevent complacency

**UX Solution**

*Implementation of an "informational countdown" and a Sensory synchrony rappresente the climb as the body perceives*

**Active Axes**

**_Emotional Risk Mediation**
*Primary, sensory synchrony*

**_Active Transparency & Cognitive Visibility**
*Secondary, maneuver motivation*

**_Active Supervision & Anti-Automation Bias**
*Activated, Vigilance micro-signals*

Framework

# Cruise

The cruise phase is the longest and can generate, depending on the profile, boredom or hyper-control

**Analytical Verifier / Control Seeking**
*Rational/Supervisor*
Over-control, Hierarchical XAI Data,
Avoid Info-overload

Control bias

**Vulnerable Passenger**
*Anxious*
Remaining Flight Time, Visible "Safe Harbor",
Positive Framing

Framing Effect

**Delegating Optimist**
*Trusting*
Surrender To Boredom, Contextual Nudges,
Disuse / Misuse Risk

UX as a regulator

## Interface Convergence

COMMON ELEMENT, ALL PROFILES
**Feedforward**
Notify before action

COMMON ELEMENT, ALL PROFILES
**On-demand Transparency Layer**
Elastic Interface (Scalable/Adaptive UI)

**Advanced XAI Menu**
A precise information hierarchy, not all available data

**Timeline**
Representing data within a positive frame: the framing effect

**Contextual Nudges**
Maintains the user in the cognitive loop, calibrated to avoid both complacency and alarmism

**UX Solution**

*Implementation of feedforward and "on-demand transparency layers", notified a few second before it occurs.*

**Active Axes**

**_Active Transparency & Cognitive Visibility**
*Primary, XIA, Feedforward, layer*

**_Emotional Risk Mediation**
*Secondary, Micro-signals of vigilance and calibration.*

**_Adaptive Trust Calibration**
*Activated, elastic interface*

Framework

# Landing & Transition

The landing phase coincides with the return to physical stability and the final confirmation of system reliability

**Analytical Verifier / Control Seeking**
*Rational/Supervisor*
Error Tolerance, Functional Specificity, Mental Model Update

**Vulnerable Passenger**
*Anxious*
Closing The Emotional Loop, Cognitive Control, Final Feedforward

**Delegating Optimist**
*Trusting*
Post-Flight Services, Flight Micro-Summary, Trust Building

## Interface Convergence

COMMON ELEMENT, ALL PROFILES
**Feedback, Final Flight Summary**
Flight completed, parameters met

COMMON ELEMENT, ALL PROFILES
**Augmented Reality (AR), Visible Vertiport**
Graphically represented landing point, physical-digital synchrony

**Landing Point + Error Tolerance**
Updates the Reliability Mental Model

**Calm Anticipatory Message**
Initial feedforword, cognitive control

**Flight Micro-summary**
Familiar summary, reinforces the model

**UX Solution**
*A final summary of the flight, reinforces the metal model for future experiences*

**Active Axes**

**_Adaptive Trust Calibration**
*Primary, Success feedback, Learned trust.*

**_Active Supervision & Anti-Automation Bias**
*Secondary, Anticipatory message, Closing the emotional loop.*

**_Active Transparency & Cognitive Visibility**
*Activated, elastic interface*

Framework

# Unexpected event

The goal of an unexpected event is to prevent a breach of trust by keeping the user in the information loop without cognitively overloading them.

**Analytical Verifier**
*Rational*
Route Discrepancy, Preventive Explanation, Critical Timing

Confirmation bias

**Control Seeking**
*Supervisor*
Loss of Agency, Perceived Control, Potential Intervention

Percepived control

**Vulnerable Passenger**
*Anxious*
Risk-As-Feeling, Salient Event, Emotional Response

Salience bias

**Delegating Optimist**
*Trusting*
Initial Trust Collapse, Cognitive Shock, Rapid Trust Erosion

Positivity bias

## Interface Convergence

COMMON ELEMENT, ALL PROFILES
**Universal Adaptive Message**
Truthiness effect, credibility through specificity

COMMON ELEMENT, ALL PROFILES
**On-demand Expansion**
Reassuring baseline for all, avoid Info-overload

**Preventive Explanation**
Critical timing: the explanation precedes the formulation of a negative hypothesis

**Visible Intervention Button**
Perceived behavioral control, intervention possible even if not necessary

**Immediate Event Normalization**
Managing the positivity bias before it turns into permanent distrust

**UX Solution**
*A model not provide separate interfaces, but a flexible and adaptive interface.*

**Active Axes**

**_Emotional Risk Mediation**
*Primary, risk-as-feeling management, salience bias.*

**_Active Transparency & Cognitive Visibility**
*Secondary, preventive explanation, expandable layer.*

**_Active Supervision & Anti-Automation Bias**
*Activated, perceived control, confident shock normalization.*

**_Adaptive Trust Calibration**
*Activated, elastic Interface, on-demand expansion for all profiles.*




Trust is not a static trait, but a dynamic interaction that takes shape within the user through external inputs